

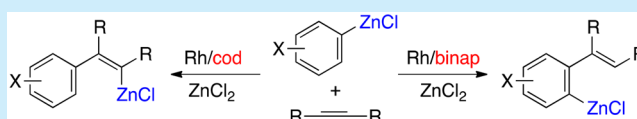
Rhodium-Catalyzed Arylzincation of Alkynes: Ligand Control of 1,4-Migration Selectivity

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Supporting Information

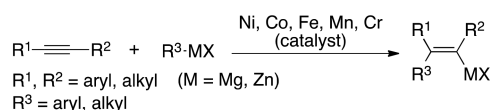
ABSTRACT: The addition of arylzinc reagents ArZnCl **1** to alkynes **2** was found to be catalyzed by rhodium complexes in the presence of a catalytic amount of zinc chloride. The selectivity in giving 2-aryllkenylzinc species **3** or *ortho*-alkenylarylzinc species **4**, the latter of which is generated through 1,4-Rh migration from alkenyl to aryl in the catalytic cycle, is controlled by the ligands on rhodium. Ligands *cod* and *binap* gave **3** and **4**, respectively, with high selectivity.



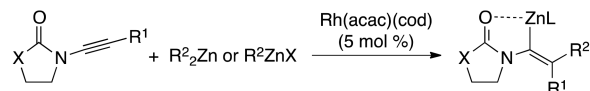
Carbometallation of alkynes is one of the most efficient methods of generating substituted alkenylmetals which are useful synthetic intermediates for the multisubstituted alkenes, and addition of arylmetals to simple unfunctionalized alkynes has been a challenging reaction because of their low reactivity.¹ The first-row transition metals including Ni, Co, Fe, Mn, and Cr have been reported to catalyze the addition of arylmagnesium² and -zinc³ reagents to the unfunctionalized alkynes (Scheme 1a). A Rh complex has been reported to catalyze the carbocyclization of ynamides⁴ (Scheme 1b), but no

Scheme 1. Addition of ArZnX to Alkynes and 1,4-Migration

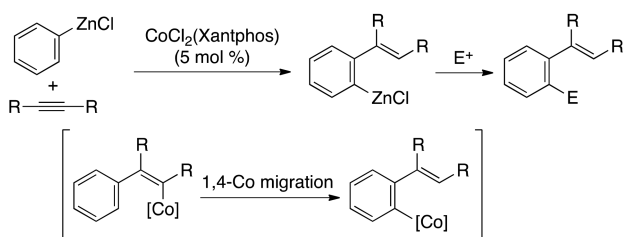
(a) Catalytic carbometallation of alkynes



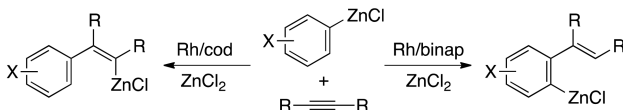
(b) Rhodium-catalyzed arylzincation of ynamides



(c) Cobalt-catalyzed addition of ArZnX to alkynes forming *ortho*-alkenylarylzincs



(d) Rh-catalyzed arylzincation. Selective in 1,4-migration or not (This Work)



reports have appeared on the carbocyclization of unfunctionalized alkynes using Rh catalysis. In 2012, Yoshikai reported⁵ a new type of Co-catalyzed arylzincation of alkynes where *ortho*-alkenylarylzinc species are formed through 1,4-migration of cobalt from an alkenyl carbon to aryl carbon (Scheme 1c). Here, we report our findings that arylzincation of unfunctionalized alkynes is catalyzed by Rh complexes in the presence of a catalytic amount of ZnCl_2 and the arylzincation with or without 1,4-migration is controlled with high selectivity by the ligands on rhodium (Scheme 1d). The 1,4-shift of rhodium from alkenyl to aryl carbons has been reported to be involved in the catalytic cycle of Rh-catalyzed multiple carbon–carbon bond forming reactions including asymmetric reactions.^{6–8} As a related arylmetalation reaction, we recently reported that the addition of arylstannanes to alkynes giving *ortho*-alkenylarylstannanes is catalyzed cooperatively by a rhodium complex and zinc chloride.⁹

In our present studies, arylzinc reagents ArZnCl were prepared by lithiation of ArBr with BuLi in THF at -80°C followed by addition of an excess amount (1.3 equiv) of ZnCl_2 to the THF solution of ArLi .^{10,11} The results obtained for the reaction of 3,5-(MeO)₂ $\text{C}_6\text{H}_3\text{ZnCl}$ (**1a**) with 4-octyne (**2a**) in the presence of several rhodium catalysts are summarized in Table 1. The reaction mixture containing 2-aryllkenylzinc **3aa** and/or *ortho*-alkenylarylzinc **4aa** was hydrolyzed with D_2O , and the location of D (**5aa/6aa**) was analyzed by ^1H and ^2H NMR spectra. The yield of the arylzincation and the selectivity with or without 1,4-migration are heavily dependent on the ligand on rhodium. Thus, the THF solution of ArZnCl (**1a**, 0.60 mmol, 3.0 equiv to **2a**) containing ZnCl_2 (0.18 mmol, 0.9 equiv to **2a**) was added to the alkyne (**2a**, 0.20 mmol) in the presence of a rhodium catalyst generated from $[\text{RhCl}(\text{coe})_2]_2$ (3 mol % of Rh) and *binap*¹² (3.3 mol %) in THF, and the mixture was heated at 50°C for 5 h. The arylzincation

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Table 1. Rhodium-Catalyzed Arylzincation of 4-Octyne (2a) with 3,5-(MeO)₂C₆H₃ZnCl (1a)^a

entry	ligand on Rh ^b	additive (mmol) ^c	yield (%) ^d 5aa + 6aa	ratio ^e of 5aa/6aa
1	binap	ZnCl ₂ (0.18)	91	<1:99
2	cod	ZnCl ₂ (0.18)	82	92:8
3 ^f	binap	ZnCl ₂ (0.09)	90	<1:99
4	binap	—	22	4:96
5 ^g	binap	—	<3	—
6 ^h	binap	ZnBr ₂ (0.18)	81	5:95
7 ^h	binap	ZnI ₂ (0.18)	87	3:97
8	dppe	ZnCl ₂ (0.18)	23	37:63
9	dppp	ZnCl ₂ (0.18)	52	48:52
10	dppf	ZnCl ₂ (0.18)	27	19:81
11	biphep	ZnCl ₂ (0.18)	83	12:88
12	PPh ₃	ZnCl ₂ (0.18)	78	6:94
13	xantphos	ZnCl ₂ (0.18)	<3	—
14	nbd	ZnCl ₂ (0.18)	83	5:95

^aReaction conditions: 4-Octyne (2a) (0.20 mmol), ArZnCl (1a) (0.60 mmol), ZnCl₂ (0.18 mmol), and Rh catalyst (3 mol % of Rh) in THF (2.0 mL) at 50 °C for 5 h. The reaction was quenched with D₂O (0.2 mL). ^b[RhCl(cod)₂]₂ (6.0 μmol of Rh) + bisphosphine (6.6 μmol), RhCl(PPh₃)₃ (6.0 μmol), or ([RhCl(diene)]₂ (diene = cod or nbd, 6 μmol of Rh). ^cThe excess amount of ZnX₂ used at the generation of ArZnX. ^dIsolated yield. ^eDetermined by ¹H and ²H NMR spectra. ^fArZnCl (1a) (0.30 mmol), ZnCl₂ (0.09 mmol), THF (1.0 mL) for 1 h. ^gReaction of Ar₂Zn generated from ArLi (0.60 mmol) and ZnCl₂ (0.30 mmol). ^hArZnX was generated from ArLi (0.60 mmol) and ZnX₂ (0.78 mmol, X = Br or I).

accompanied by the 1,4-migration proceeded selectively to give, after the D₂O quenching, a 91% yield of 6aa, where the deuterium is incorporated at the *ortho*-position of the phenyl ring (entry 1). On the other hand, the reaction in the presence of a rhodium catalyst coordinated with cod¹² under otherwise the same conditions gave an 82% yield of the arylation product where the deuterium is found mainly at the alkenyl carbon (5aa/6aa = 92:8) (entry 2). In the Rh/binap-catalyzed reaction, the amounts of ArZnCl 1a and ZnCl₂ were reduced to 1.5 and 0.45 equiv, respectively, without loss of the arylzincation yield or 1,4-migration selectivity (entry 3). The presence of ZnCl₂ as a catalyst, which was added in an excess amount (1.3 equiv to ArLi) at the generation of ArZnCl from ArLi, is essential for the present arylzincation. The yield of arylzincation was much lower (22%) in the reaction with ArZnCl generated with 1.0 equiv of ZnCl₂ (entry 4). Diarylzinc Ar₂Zn generated from ArLi and 0.50 equiv of ZnCl₂ did not give the arylzincation products (entry 5). Other zinc salts, ZnX₂ (X = Br and I), which were used in an excess amount at the generation of ArZnX¹³ from ArLi, also catalyzed the reaction, but the yields and the migration selectivity were somewhat lower (entries 6 and 7). The lower yield and selectivity were observed with other phosphine ligands, dppe, dppp, dppf, biphep, and PPh₃ (entries 8–12). The arylzincation did not proceed with the xantphos ligand (entry 13), which has been reported to be the best ligand in the cobalt-catalyzed migratory arylzincation.⁵ Interestingly, the high selectivity in giving 3aa without the 1,4-migration was observed only with 1,5-cyclooctadiene (cod). Other diene ligands including norbornadiene (nbd) promoted the migratory arylzincation, giving 4aa preferentially (entry 14).

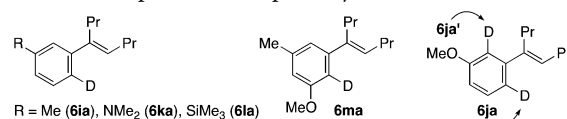
The rhodium-catalyzed arylzincation was also successful with several other arylzinc reagents ArZnCl, and similar

selectivity control in giving arylzincation products with or without 1,4-migration was observed in the arylzincation of 4-octyne (2a), whose results are summarized in Table 2. The

Table 2. Rhodium-Catalyzed Arylzincation of 4-Octyne (2a) with Arylzinc Reagents 1^a

entry	1: Ar in ArZnCl	L on Rh ^b (A) or (B) ^a	yield (%) ^c of 5 + 6	ratio ^d of 5:6
1	1a: 3,5-(MeO) ₂ C ₆ H ₃	binap (B) cod (A)	90 82	<1:99 92:8
2	1b: Ph	binap (A) cod (A)	87 80	2:98 80:20
3	1c: 4-MeC ₆ H ₄	binap (B) cod (A)	91 68	1:99 77:23
4	1d: 4-MeOC ₆ H ₄	binap (A) cod (A)	94 84	1:99 74:26
5	1e: 4-Me ₂ NC ₆ H ₄	binap (B) cod (A)	87 71	1:99 94:6
6	1f: 4-Me ₃ SiC ₆ H ₄	binap (B) cod (A)	85 86	6:94 81:19
7	1g: 4-FC ₆ H ₄	binap (B) cod (A)	83 52	<1:99 49:51
8	1h: 4-CF ₃ C ₆ H ₄	binap (B) cod (A)	93 87	<1:99 <1:99
9	1i: 3-MeC ₆ H ₄	binap (B) cod (A)	87 72	1:99 ^e 86:14 ^e
10	1j: 3-MeOC ₆ H ₄	binap (A) cod (A)	93 82	1:99 ^f 80:20 ^f
11	1k: 3-Me ₂ NC ₆ H ₄	binap (B) cod (A)	95 72	1:99 ^e 95:5 ^e
12	1l: 3-Me ₃ SiC ₆ H ₄	binap (B) cod (A)	87 79	1:99 ^e 83:17 ^e
13	1m: 3-Me-5-MeOC ₆ H ₃	binap (B) cod (A)	97 79	1:99 ^e >99:1
14	1n: 2-MeOC ₆ H ₄	binap (A) cod (A)	88 64	13:87 94:6

^aReaction condition (A): ArZnCl 1 (0.60 mmol), ZnCl₂ (0.18 mmol), 4-octyne (2a) (0.20 mmol), and Rh catalyst (3 mol % of Rh), THF (total 2.0 mL) at 50 °C for 5 h. The reaction was quenched with D₂O (0.2 mL). (B): ArZnCl 1 (0.30 mmol), ZnCl₂ (0.09 mmol), 4-octyne (2a) (0.20 mmol), and Rh catalyst (3 mol % of Rh), THF (total 1.0 mL) at 50 °C for 1 h. ^b[RhCl(cod)₂]₂ (6 μmol of Rh) + binap (6.6 μmol) or [RhCl(cod)]₂ (6 μmol of Rh). ^cIsolated yield. ^dDetermined by ¹H and ²H NMR. ^eRegioselective 1,4-shift giving the products 6 shown below. ^fA mixture of 6ja and 6ja' in a ratio of 2:1 and 4:1 with binap and cod, respectively.

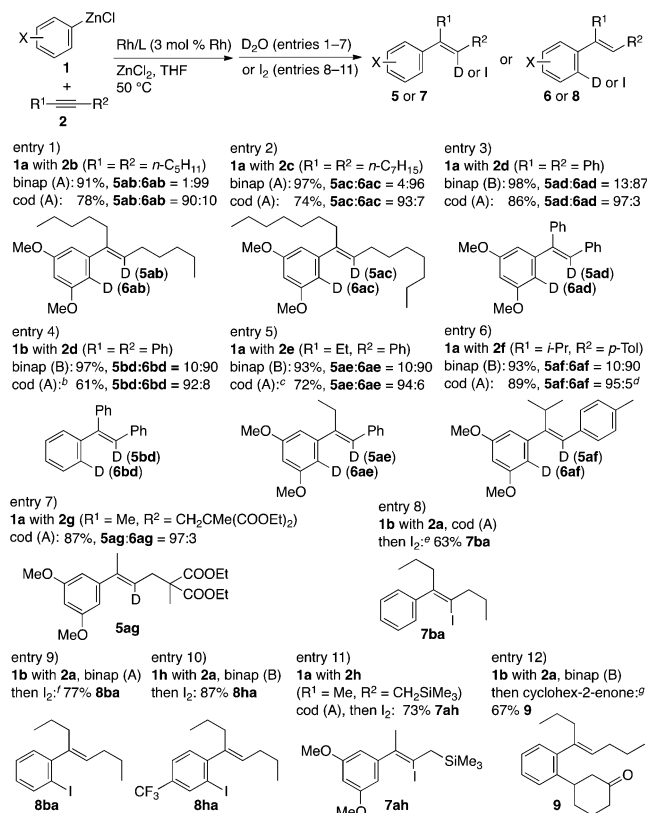


selectivity of the addition/1,4-migration sequence giving *ortho*-alkenylphenylzinc 4 with binap as a ligand is very high for all the arylzinc reagents shown in Table 2, including unsubstituted phenylzinc (entry 2) and *para*-substituted ones with both electron-donating and -withdrawing groups (entries 3–8). The ratio of 5:6 ranges between <1:99 and 6:94. The 1,4-shift was regioselective in the reaction of *meta*-substituted phenylzinc reagents with Me, NMe₂, and SiMe₃, deuterium being incorporated at the less hindered position exclusively (entries 9, 11, 12). The regioselectivity was low with 3-MeOC₆H₄ZnCl (entry 10). Interestingly, the 1,4-migration took place selectively to the position next to the MeO group in the

reaction of 3-Me-5-MeOC₆H₃ZnCl (entry 13). The selective formation of 2-arylethenylzincs **3** without 1,4-migration was also realized using cod as a ligand of the rhodium catalyst for most of the arylzinc reagents examined, although the selectivity is generally not as high as that for the reaction with migration. The ratio of **5**:**6** after D₂O quenching is higher than 7:3, except for the arylzinc reagents substituted with electron-withdrawing groups. Zinc reagents 4-FC₆H₄ZnCl and 4-CF₃C₆H₄ZnCl gave the corresponding arylation products with **5**:**6** = 49:51 and <1:99, respectively¹⁴ (entries 7 and 8).

The reaction conditions used for the arylzincation of 4-octyne are applicable to other unfunctionalized alkynes. The results are summarized in Scheme 2, which also contains

Scheme 2. Rhodium-Catalyzed Arylzincation of Alkynes 2 with Arylzinc Reagents 1^a



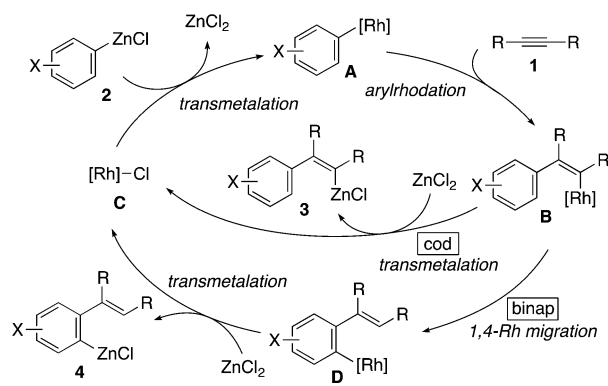
^aFor reaction conditions (A) and (B), see footnote a in Table 2. For iodination, I₂ (0.60 mmol) was added after the Rh-catalyzed arylzincation. ^bTHF solutions of [RhCl(cod)]₂ and alkyne **2d** were added dropwise to PhZnCl (**1b**) and ZnCl₂ in THF over 40 min. ^c1.5 mol % of Rh catalyst. ^dContaining 5% of its regioisomer. ^e5 mmol scale. ^f20 mmol scale, 1 mol % of Rh catalyst. ^gCyclohex-2-enone (0.36 mmol) and [RhCl(cod)]₂ (15 mol % Rh) were added, and the mixture was heated at THF reflux for 12 h.

examples where the generated zinc reagents were used for a few transformations other than the deuteration.¹⁵ The arylzincation with **1a** took place for symmetrically substituted dialkylalkynes **2b–2c** and diphenylacetylene (**2d**) to give the corresponding *ortho*-alkenylarylzinc or 2-arylethenylzinc products with high selectivity using binap or cod (entries 1–4). In the reactions of alkyl(aryl)alkynes **2e–2f**, high regioselectivity was observed for the bond formation between the aryl group of the arylzinc and the alkyl-substituted alkyne carbon (entries 5 and 6). This selectivity is as expected from the reported

regiochemistry at carbometalation of alkyl(aryl)alkynes.¹ The high regioselectivity was also observed in the arylzincation of methyl(alkyl)alkyne **2g**, where the alkyl group is much bigger than methyl (entry 7). The iodination of arylzinc reagents, 2-arylethenylzinc **3** and *ortho*-alkenylarylzinc **4**, with I₂ proceeded smoothly to give the corresponding alkenyl and aryl iodides in high yields (entries 8–11). As another example of transformation of the resulting zinc reagents, Rh-catalyzed conjugate addition of the arylzinc species¹⁶ generated from 4-octyne (**2a**) and PhZnCl (**1b**) is shown in entry 12.¹⁷

The catalytic cycle of the present arylzincation of alkynes with/without 1,4-migratory is proposed as shown in Scheme 3.

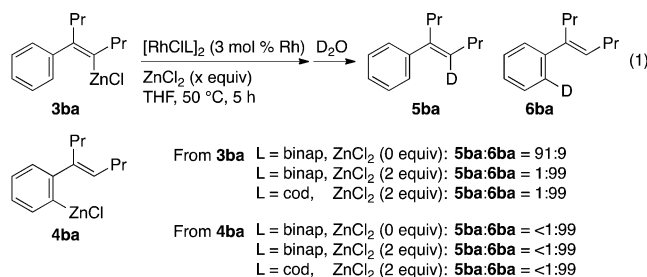
Scheme 3. A Catalytic Cycle Proposed for Arylzincation of Alkynes Catalyzed by Rh Complexes and ZnCl₂



Thus, the syn-addition of aryl–Rh intermediate **A** to the alkyne generates 2-arylethenyl–Rh species **B**. Transmetalation of the alkenyl group from Rh to ZnCl₂ takes place to produce alkenylzinc product **3** and Cl–Rh **C**. Arylation of **C** with ArZnCl regenerates aryl–Rh **A**.^{18,19} Direct transmetalation between alkenyl–Rh **B** and ArZnCl **2** giving aryl–Rh **A** and alkenylzinc **3** is less likely because a catalytic amount of ZnCl₂ is necessary for the present arylzincation to take place (see entries 4 and 5 in Table 1). When the transmetalation of alkenyl–Rh intermediate **B** with ZnCl₂ is relatively slow, it undergoes 1,4-migration of Rh giving *ortho*-alkenylphenyl–Rh intermediate **D**, which is thermodynamically more stable than **B**.^{7e} Transmetalation with ZnCl₂ finally leads to the *ortho*-alkenylarylzinc **4**.²⁰ The selectivity in producing **3** or **4** is dependent on the reactivity of alkenyl–Rh intermediate **B** toward transmetalation with ZnCl₂. It is our understanding that the transmetalation, which is an intermolecular reaction, is faster with a smaller cod ligand and the Rh 1,4-migration which is an intramolecular event is not strongly affected by the ligands on rhodium.

The reactions starting with alkenylzinc **3ba** and *ortho*-alkenylarylzinc **4ba** (eq 1) gave us further insight into the reaction mechanism. The isomerization of **3ba** into **4ba** was observed in the presence of ZnCl₂ with both Rh/binap and Rh/cod catalysts, while the isomerization is very slow without ZnCl₂. The isomerization from **4ba** to **3ba** was not observed under the same conditions with the binap or cod ligand. These results demonstrate that ZnCl₂ plays a key role at the transmetalation steps in the catalytic cycle and that *ortho*-alkenylarylzinc **4ba** is thermodynamically more stable than alkenylzinc **3ba**. It is remarkable that the ratio of 2-arylethenylzinc **3** to *ortho*-alkenylarylzinc **4** in the reaction of alkyne **2** with arylzinc **1** is not dependent on the reaction time

or conversion. Thus, D₂O quench of the reaction of **1a** with **2a** at 15 min reaction time gave a 47% yield of **5aa/6aa** (1/99) and a 29% yield of **5aa/6aa** (93/7) with binap and cod ligands, respectively. The **5aa/6aa** ratios are essentially the same as those observed in entries 1 and 2 in Table 1. The organozincs **3** and **4** generated by the reaction of alkyne **2** with an excess amount of ArZnCl **1** do not go back to the catalytic cycle shown in Scheme 2 in the presence of excess **1**, although both Rh/binap and Rh/cod catalyze the isomerization of **3** to **4** in the absence of ArZnCl **1** (eq 1).



In summary, we have disclosed that the addition of arylzinc reagents ArZnCl to unfunctionalized alkynes is efficiently catalyzed by rhodium complexes in the presence of a catalytic amount of zinc chloride. One of the divergent pathways in giving 2-arylalkenylzincs or *ortho*-alkenylarylzincs, the latter of which are generated through 1,4-Rh migration from alkenyl to aryl in the catalytic cycle, is chosen by use of the appropriate ligands on rhodium.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.8b02668.

Experimental procedures, compound characterization data, and crystallographic data (PDF)

Accession Codes

CCDC 1813165 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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(18) Transmetalation of Ar groups from ArZnX to Rh(I) has been reported in Rh-catalyzed conjugate arylation with ArZnX (ref 15).

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(20) The transmetalation between Ar-Rh and ZnCl_2 giving ArZnCl has been proposed in the migratory arylstannylation catalyzed by a Rh catalyst and ZnCl_2 . See ref 9.